

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
## $n.burnin
## [1] 1000
##
## $n.draw
## [1] 1200
##
## $thin
## [1] 1
##
## $pp_cutoff
## [1] 0.5
##
## $prior_gamma
## [1] 0.1 0.1 0.1 0.1 0.1
##
## $K_values
## [1] "1 / Examined number of clusters: 2" "2 / Examined number of clusters: 3"
## [3] "3 / Examined number of clusters: 4" "4 / Examined number of clusters: 5"
## [5] "5 / Examined number of clusters: 6" "6 / Examined number of clusters: 7"
## [7] "7 / Examined number of clusters: 8" "8 / Examined number of clusters: 9"
## [9] "9 / Examined number of clusters: 10"
##
## $sdev
## [1] 0.01
##
## $beta.var.scale
## [1] 0.8
```

Summary Statistics

```
--- Summary for K = 1 ---
Z.ar (Location: fit[[1]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.6658  0.7050  0.7142  0.7144  0.7242  0.7617

beta.ar (Location: fit[[1]]$beta.ar):
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.09104 0.14167 0.13509 0.17500 0.93167
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Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[1]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0950 0.1450 0.1384 0.1783 0.7725
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.101739 0.030833 1.000000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.076899 0.024167 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005833 0.015000 0.162343 0.119167 1.000000

--- Summary for K = 2 ---
Z.ar (Location: fit[[2]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.5192 0.5758 0.5867 0.5866 0.5975 0.6308

beta.ar (Location: fit[[2]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.04667 0.09833 0.10406 0.14333 0.83250
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[2]]$gamma.ar):
  Modality 1 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.04833	0.10167	0.10676	0.14833	0.69583

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000000	0.0008333	0.0383333	0.4023191	1.0000000	1.0000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.003333	0.239880	0.273958	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	0.085	1.000	0.685	1.000	1.000

--- Summary for K = 3 ---

Z.ar (Location: fit[[3]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4433	0.4833	0.4950	0.4945	0.5042	0.5517

beta.ar (Location: fit[[3]]\$beta.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.02917	0.06917	0.07978	0.11167	0.83333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar (Location: fit[[3]]\$gamma.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.03167	0.07250	0.08249	0.11667	0.63917

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1525	0.5008	0.5341	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2555	0.5360	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3233	1.0000	0.6927	1.0000	1.0000

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--- Summary for K = 4 ---
Z.ar (Location: fit[[4]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.4833 0.5433 0.5708 0.5717 0.5967 0.6850

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beta.ar (Location: fit[[4]]$beta.ar):
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.02000 0.05000 0.06625 0.09000 0.78333
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1

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gamma.ar (Location: fit[[4]]$gamma.ar):
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.02167 0.05167 0.06737 0.09333 0.51667
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.07970 0.05083 1.00000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.8608 0.9200 0.9063 0.9717 1.0000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.04139 0.00000 0.76750
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.1167 0.1892 0.3558 1.0000

```

```

--- Summary for K = 5 ---
Z.ar (Location: fit[[5]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.4267 0.4700 0.4825 0.4833 0.4975 0.5500

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beta.ar (Location: fit[[5]]$beta.ar):
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01333 0.04167 0.05821 0.08167 0.81000
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1

```

```

Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1

gamma.ar (Location: fit[[5]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01500 0.04500 0.05993 0.08667 0.73500
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0000  0.0000  0.1619  0.2233  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.1233  1.0000  1.0000  0.9975  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.09165 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.03833 0.40583 0.41390 0.67417 1.00000

--- Summary for K = 6 ---
Z.ar (Location: fit[[6]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.8600  0.9808  0.9842  0.9839  0.9883  0.9967

beta.ar (Location: fit[[6]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01167 0.02917 0.05166 0.06333 0.86583
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1

gamma.ar (Location: fit[[6]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01167 0.03167 0.05130 0.06667 0.67500
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0001523 0.0000000 1.0000000
  Modality 3 :

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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0e+00	0.0e+00	0.0e+00	7.6e-05	0.0e+00	1.0e+00

--- Summary for K = 7 ---

Z.ar (Location: fit[[7]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8083	0.9100	0.9283	0.9237	0.9417	0.9725

beta.ar (Location: fit[[7]]\$beta.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008333	0.024167	0.053403	0.057500	0.867500

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar (Location: fit[[7]]\$gamma.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01000	0.02500	0.04944	0.06167	0.64417

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.001408	0.000000	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0000347	0.0000000	0.6333333

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.01365	0.00000	1.00000

--- Summary for K = 8 ---

Z.ar (Location: fit[[8]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8517	0.9800	0.9850	0.9837	0.9883	0.9958

beta.ar (Location: fit[[8]]\$beta.ar):

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00750 0.01917 0.04327 0.04667 0.85750
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1         1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1         1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1         1         1         1         1         1

gamma.ar (Location: fit[[8]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.008333 0.021667 0.041299 0.050000 0.533333
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00e+00 0.00e+00 0.00e+00 3.53e-05 0.00e+00 1.00e+00
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0         0         0         0         0         0
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0         0         0         0         0         0
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0         0         0         0         0         0

--- Summary for K = 9 ---
Z.ar (Location: fit[[9]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.8108  0.9067  0.9200  0.9168  0.9300  0.9567

beta.ar (Location: fit[[9]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.005833 0.014167 0.029817 0.034167 0.835000
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1         1         1         1         1         1

```

```

gamma.ar (Location: fit[[9]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.006667 0.015000 0.030245 0.036667 0.483333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.000287 0.000000 1.000000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.03605 0.000000 1.000000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
      0      0      0      0      0      0
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000000 0.0000000 0.0000000 0.0002176 0.0000000 1.0000000

```